

TIC 111149941

Eclipsing binary candidate dossier - Star Vetter

AMBIGUOUS PHOTOMETRY

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	5.240092 d	VSX	no match
Epoch (t0)	2695.2575 BTJD	ASAS-SN	no match
Duration	0.3500 d	Gaia eclipsing-binary table	no match
Depth	0.045 % (455 ppm)	Gaia variable classification	no match
SNR	10.2	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	3 (gate >= 3)	Verdict:	NOVEL
Robust depth ratio	1.02 (gate >= 0.50)		
Robust depth sigma	8.5 (gate >= 7.0)		
Odd/even z-score	2.29 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

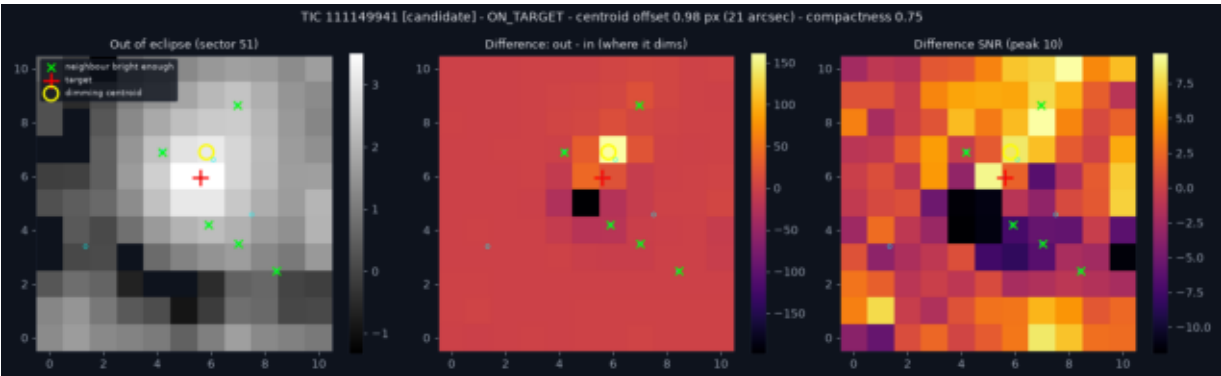
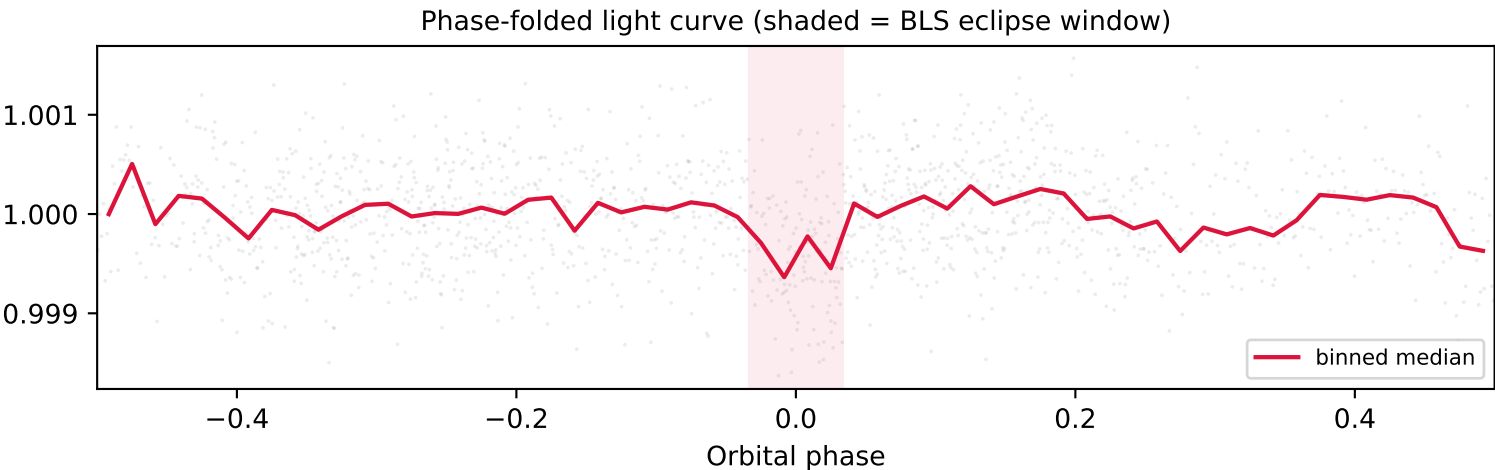
Sector 51 light-curve dip in this sector: 7.9 sigma t0 refined by 0.0100 phase

Difference-image peak SNR 9.73 (gate >= 5.0) Compactness 0.75 (gate >= 0.50)

Dimming centroid offset 0.98 px (21 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 1002359509

1.65 px away, dTmag +8.06 - farther from the dimming than the target



Caveat (AMBIGUOUS PHOTOMETRY): The difference-image detection is strong (peak SNR 9.7, compactness 0.75) but the star's own simple-aperture photometry shows no dip, and aperture/quality-mask sensitivity tests were inconsistent. Confirmed by pixel imaging; not independently confirmed by simple photometry.